EXECUTIVE ARCHITECTURE BRIEFING

Alvi Quantitative Terminal & Autonomous AI Engine

Institutional-grade swing execution engine integrating AAOIFI Shariah compliance, dual-LLM quantitative reasoning, and Groww Delivery API execution.

100% AAOIFI Shariah

AAOIFI

ETHICAL ISLAMIC ASSET SCREEN

- **Strict 0% Riba:** Zero conventional banks, liquor, gambling, tobacco or weapons.
- **Debt / Mcap <= 33%:** Strictly eliminates over-leveraged companies operating on interest.
- **Cash / Mcap <= 33%:** Ensures capital backs productive real enterprise value.
- **Delivery CNC Only:** Actual shares owned in Demat; zero margin or short selling.

Mathematical Rigor

DONCHIAN

20-DAY DONCHIAN BREAKOUTS

- **Channel Breakouts:** Clean clearance over 20-day high with volatility expansion.
- **Trend Alignment:** Price > 50 SMA > 200 SMA (Golden Trend Regime stack).
- **RSI Momentum:** Optimal 55–70 band; rejects parabolic moves (>75 RSI).
- **Asymmetric Payoff:** Strict 2:1 risk/reward target (+8% Target vs -4% Stop).

Dual AI Brain

DUAL LLM

CLAUDE + GEMINI ENGINE

- **Claude Sonnet 5.5:** Primary risk & conviction analyst (>=7.5 / 10 threshold).
- **Gemini 3.5 Flash:** Standby fallback engine with automatic failover circuit.
- **Beyond Numbers:** Evaluates management changes, catalysts & crowd psychology.
- **Prepaid Budget:** Real-time token cost tracking against prepaid $5.00 limit.

Production Cluste

DOCKER :14

CAPROVER ON ORACLE CLOUD

- **Daily Cron (16:00 IST):** Autonomous post-market analysis & Groww order routing.
- **Mid-Day Watchdog:** 120s intraday stop-loss (-4%) & profit trailing loop.
- **Anti-Throttling:** RAM cache + Google Finance fallback on cloud IPs.
- **Live Terminal:** OLED Dark Web UI with HMAC-SHA256 session cookies.

STAGE 1: ALGORITHMIC SELECTION

Pre-Screening Engine & Technical Indicators Checked

Thousands of NSE cash stocks are mathematically evaluated before any AI tokens are consumed, filtering for high-probability momentum setups.

1. 20-Day Donchian Breakout

BREAKOUT

VOLATILITY & RESISTANCE CLEARANCE

- **Primary Formula:** Close > High_20d with expanding multi-timeframe volume.
- **Clearance Margin:** Requires >0.5% clearance over the 20-day high to eliminate false wicks and traps.
- **Institutional Base:** Identifies multi-week accumulation transitioning into an explosive markup trend.
- **Capital Price Cap:** Filters equities <= Rs 5,000 to guarantee exact CNC position sizing within allocation limits.

2. Moving Average Alignment

MA STACK

MULTI-TIMEFRAME TREND STACK

- **The Golden Stack:** Enforces Price > 50-day SMA > 200-day SMA in strict ascending hierarchy.
- **Long-Term Filter:** Disqualifies stocks below 200 SMA (such as Patanjali at 411.5 vs 422.5 SMA200) regardless of news hype.
- **Base Slope:** Demands an upward trajectory on the 50 SMA to avoid buying declining structural bear moves.
- **Extension Guard:** Rejects stocks extended >25% above 50 SMA to avoid buying at exhaustion points.

3. RSI & Momentum Velocity

OSCILLATOR

NON-EXHAUSTION & VELOCITY AUDITING

- **14-Period RSI:** Target zone is strictly 55.0 to 70.0 (sweetspot of established momentum with upside room).
- **Overbought Veto:** Hard rejection on parabolic profiles with RSI > 75-80 (e.g. SAILIFE at 93.4, INDHOTEL at 84.0).
- **5d & 20d Velocity:** Cross-checks 5-day return (+5% to +12%) with 20-day trend to confirm accelerating volume.
- **Stall Guard:** Rejects stocks showing negative 5-day momentum on 20-day highs (e.g. LTTS stalled at -0.5% 5d).

STAGE 2: SYNTHETIC INTELLIGENCE

The AI Quantitative Brain: Qualitative Multi-Factor Reasoning

Raw technicals only provide candidates. The LLM Brain synthesizes balance sheet health, trader psychology, macro tape, and corporate catalysts.

Fundamental Durability & Capital Health

BALANCE SHEET

P&L, DEBT BURDEN & RETURN ON EQUITY (ROE)

- **Return on Equity:** Prioritizes efficient capital allocators (ROE >= 15%). Rejects weak/negative ROE (e.g. Ather at -6.7%).
- **Leverage Health:** Inspects Debt-to-Equity & interest coverage to ensure survival through market downturns.
- **Operating Margin:** Evaluates quarterly EBITDA expansion vs commodity input price compression.

Market Regime & Tape Context

REGIME

BULLISH VS NEUTRAL VS DEFENSIVE REGIME

- **Macro Tape:** Diagnoses whether the market tape is BULLISH, NEUTRAL, or DEFENSIVE using Nifty 50 breadth.
- **Risk Expectancy:** In a NEUTRAL tape, tight 4% stops on extended runs have negative expected value.
- **Discipline:** Claude elected PASS on all 10 candidates in neutral tape, preserving 100% capital from pullback traps.

Trader Mindset & Crowd Dynamics

PSYCHOLOGY

ACCUMULATION VS RETAIL FOMO & GAP RISK

- **Smart Money:** Distinguishes early institutional accumulation from late retail FOMO buying at the peak of a 30% surge.
- **Overnight Risk:** Evaluates vulnerability to weekend headline gap-downs and post-earnings hangover.
- **Tactical Retest:** Instructs waiting for breakout retest (e.g. Shyam Metalics ~1,270) rather than market-order chasing.

Corporate Catalysts & Sizing Conviction

CATALYSTS

MATERIAL INFLECTION POINTS & SIZING

- **Leadership Turnaround:** Ingests executive changes (e.g. incoming CEO re-rating prospects).
- **Sector Tailwinds:** Incorporates national infrastructure spending, metals demand, or export policy shifts.
- **Conviction Scoring:** Grades 1.0 to 10.0 scale; requires >=7.5 with 2:1 profit asymmetry (+8% target / -4% stop) to execute.

STAGE 3: AUTONOMOUS EXECUTION

Autonomous Execution & Intraday Protection

End-to-end bridge between AI strategic verdicts and live market orders on Groww, guarded by real-time mid-day stop loss and trailing profit loops.

Daily Execution Engine

CRON 16:00

AGENT-JOBS (16:00 IST MON–FRI)

- **Post-Market Loop:** Wakes at 16:00 IST after exchange closes to analyze verified daily settlement bars.
- **Capital Sizing:** Dynamically calculates shares based on live Groww cash balance, capped at Rs 5,000 per position.
- **Groww API Routing:** Directly submits orders to Groww API as PRODUCT_CNC (Cash Delivery) using automated TOTP 2FA.
- **State Sync:** Persists execution signals, target levels, and stop prices to data/ai_signals.json.

Mid-Day Price Guardian

WATCHDOG

AGENT-MONITOR (09:15–15:30 IST)

- **Intraday Watchdog:** Watches live Groww holdings every 120 seconds throughout NSE open market hours.
- **Stop-Loss (-4%):** If price drops <= stop price mid-day, verifies Kill Switch and executes an immediate MARKET CNC SELL on Groww.
- **Target Hit (+8%):** Triggers an instant mid-day query to the AI Brain for strategic momentum evaluation.
- **Zero Night Calls:** Completely sleeps during nights and weekends, ensuring 0 unnecessary API requests.

AI Profit-Maximizer

TRAILING

ADAPTIVE PROFIT TRAILING ADVISOR

- **The Strategic Choice:** Does the winning stock have legs to run +15%, or is momentum exhausting at +8%?
- **If TAKE_PROFIT:** Orders an immediate CNC SELL to lock in the +8% gain into Demat.
- **If HOLD_AND_TRAIL:** Raises target to +12%/+14% and raises stop loss to a trailing +5% locked-in profit in state.
- **Wave Capture:** Allows riding explosive multi-week momentum without cutting winning trades short.

ENTERPRISE RISK CONTROLS

Multi-Layer Risk Architecture & Cloud Resilience

Built for zero-downtime autonomy on Oracle Cloud: protecting cloud IP addresses, budget safeguards, automatic LLM failover, and universal kill switch.

Data Center Anti-Throttling Guard

ANTI-THROTTLE

ORACLE CLOUD IP RATE-LIMIT DEFENSES

- **In-Memory 30s Cache:** Repeated quote queries for the same symbol are served instantly from RAM with 0 network calls.
- **Google Finance Fallback:** If Yahoo returns 429 Too Many Requests or blocks Oracle IPs, Google Finance seamlessly takes over.
- **120s Polling Cadence:** Only ~187 quote requests over the entire 6.25h trading day, well below all rate limits.

Prepaid AI Budget & Dual-LLM Failover

FAILOVER

ZERO INTERRUPTION & SPEND GOVERNANCE

- **Prepaid $5.00 Cap:** Real-time token cost tracking ($0.0795 spent / $4.9205 remaining across 7 calls).
- **Auto Gemini Takeover:** If Claude budget exhausts or Anthropic errors, Google Gemini automatically executes trades.
- **Dual Visibility:** Dashboard displays real-time spend, call counts, and failover status for both Claude & Gemini.

Physical Emergency Kill Switch

KILL SWITCH

UNIVERSAL VETO AUTHORITY

- **Instant Circuit Breaker:** Engaging Kill Switch instantly blocks all buy and sell orders across all containers.
- **Single-Click UI:** Prominent emergency halt button on the live OLED dashboard at trading.lifemint.net.
- **Persistent State:** Recorded in data/kill_switch.json with ISO-8601 audit timestamp and trigger reason.

Production Cluster & Access Security

CAPROVER :14

CLEAN DEPLOYMENT ARCHITECTURE

- **Docker Swarm Cluster:** Running live on CapRover Oracle Cloud host (Image Tag :14).
- **Zero Duplication:** Strictly audited single-source environment variables across all services.
- **HMAC-SHA256 Auth:** Tamper-evident 30-day session cookies with timing-safe comparison on custom username.